Answers learning task 5

2026 – 2027

Exercise 4.1: Suppose that $f(x) = 2e^{-2x}$ for $0 < x$. Determine the following:

(a) $P(1 < X)$
(b) $P(1 < X < 2.5)$
(c) $P(X = 3)$
(d) $P(X < 4)$
(e) $P(3 \leq X)$
(f) Determine x such that $P(x < X) = 0.20$.
(g) Determine x such that $P(X \leq x) = 0.10$.

Answer

(a) 0.1353
(b) 0.1286
(c) 0
(d) 0.9997
(e) 0.0025
(f) 0.8047
(g) 0.0527

Exercise 4.7: Suppose that $f(x) = \frac{3}{2}x^2$ for $-1 < x < 1$. Determine the following:

(a) $P(0 < X)$
(b) $P(0.5 < X)$
(c) $P(-0.5 \leq X \leq 0.5)$
(d) $P(X < -2)$
(e) $P(X < 0 \text{ or } X > -0.5)$
(f) x such that $P(x < X) = 0.05$.

Answer

(a) 0.5
(b) 0.4375
(c) 0.125
(d) 0
(e) 1
(f) 0.9655

Exercise 4.11: The probability density function of the length of a metal rod is $f(x) = 2$ for $2.3 < x < 2.8$ meters.

(a) If the specifications for this process are from 2.25 to 2.75 meters, what proportion of rods fail to meet the specifications?

(b) Assume that the probability density function is $f(x) = 2$ for an interval of length 0.5 meters. Over what value should the density be centered to achieve the greatest proportion of rods within specifications?

Answer

(a) 0.1

(b) 2.5, thus $f(x) = 2$ for $2.25 < x < 2.75$

Exercise 4.13: Suppose that the cumulative distribution function of the random variable X

is

$$F(x) = \begin{cases} 0 & x < 0 \\ 0.2x & 0 \le x < 5 \\ 1 & 5 \le x \end{cases}$$

Determine the following:

(a) $P(X < 2.8)$
(b) $P(X > 1.5)$
(c) $P(X < -2)$
(d) $P(X > 6)$

Answer

(a) 0.56
(b) 0.7
(c) 0
(d) 0

Exercise 4.15: Determine the cumulative distribution function given the following probability density function:

$$f(x) = 2e^{-2x} \qquad \text{for } 0 < x$$

Answer

$$F(x) = \begin{cases} 0 & x \le 0 \\ 1 - e^{-2x} & x \ge 0 \end{cases}$$

Exercise 4.21: Determine the cumulative distribution function for the following probability density function:

$$f(x) = \begin{cases} 2, & \text{for } 2.3 < x < 2.8 \\ 0, & \text{otherwise.} \end{cases}$$

Use the cumulative distribution function to determine the probability that a length exceeds 2.7 meters.

Answer

$$F(x) = \begin{cases} 0 & x \le 2.3 \\ 2x - 4.6 & 2.3 \le x \le 2.8 \\ 1 & x \ge 2.8 \end{cases}$$

$$P(X > 2.7) = 0.2$$

Exercise 4.29: Suppose that $f(x) = \frac{3}{2}x^2$ for $-1 < x < 1$. Determine the mean and variance of X.

Answer

$E(X) = 0, \quad V(X) = 0.6$

Exercise 4.37: Integration by parts is required. The probability density function of the diameter of a drilled hole in millimeters is $f(x) = 10e^{-10(x-5)}$ for $x > 5$ mm. Although the target diameter is 5 millimeters, vibrations, tool wear, and other nuisances produce diameters greater than 5 millimeters.

(a) Determine the mean and variance of the diameter of the holes.

(b) Determine the probability that a diameter exceeds 5.1 millimeters.

Answer

(a) $E(X) = 5.1, \quad V(X) = 0.01$

(b) 0.3679

Exercise 4.41: The thickness of a flange on an aircraft component is uniformly distributed between 0.90 and 1.05 millimeters.

(a) Determine the cumulative distribution function of flange thickness.

(b) Determine the proportion of flanges that exceeds 1.02 millimeters.

(c) What thickness is exceeded by $90\,\%$ of the flanges?

(d) Determine the mean and variance of flange thickness.

Answer

(a)
$$F(x) = \frac{20}{3}x - 6 \quad \text{for } 0.9 \leq x \leq 1.05$$

(b) 0.2

(c) 0.915

(d) $E(X) = 0.975, \quad V(X) = 0.001875$